

A Unified Mathematical Model for Truth-Probability and Importance-Scoring of Statements

Mohamed Amine Belachhab

Independent Researcher, Fes, Morocco

m.amine.belachhab@gmail.com

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Abstract

This paper presents a unified mathematical framework for assigning two scores to any statement s made by a source to a receiver r : an evidence-weighted truth-probability score and an importance score. The framework produces three outputs — $T_{\text{abs}}(s)$, an evidence-weighted truth-probability computed as if evaluated by a highly unbiased and fairly skeptical expert; $T_{\text{perc}}(s,r)$, the perceived truth-probability as modulated by delivery, context, and receiver properties; and $I(s,r)$, the importance score, which is always receiver-dependent. The central architectural claim is that the input space of T_{abs} is a proper subset of the input space of T_{perc} , which is a proper subset of the input space of I . All inter-cluster updates operate via sequential Bayesian updating with a likelihood ratio function derived from cluster-specific neutral points. The framework integrates six structured input clusters covering source credibility, statement properties, delivery context, situational context, receiver properties, and corroboration. It also introduces a relevance-gated situational context mechanism, $\text{Rel}(s,c)$, which distinguishes ambient contextual stakes from statement-specific relevance to those stakes. Fifteen formal axioms ground the architectural decisions. Sixteen open problems are identified and precisely stated. Fifteen stress-test cases are evaluated in full.

Keywords: truth-probability, importance scoring, Bayesian epistemology, source credibility, statement evaluation, information quality, relevance gating

1 Introduction

The problem of assigning formal truth-probabilities to statements has been approached from several directions: Bayesian epistemology treats belief as subjective probability updated on evidence; formal pragmatics examines how context modulates meaning; source credibility research identifies trust and expertise as primary credibility dimensions. What is less developed is a single unified framework that formally separates evidence-weighted from perceived truth-probability, treats both as projections of a richer importance space, and provides a computationally tractable scoring architecture grounded in information theory and Bayesian inference.

This framework attempts to address that gap. It is both normative — specifying how truth and importance should be weighted given rational epistemic standards — and empirically tractable, drawing on observable inputs organized into six structured clusters. The empirical parameters it introduces are declared explicitly and their validation is reserved for a sequel study applying the framework to benchmark datasets through a human-aligned variable-rubric protocol, evidence audit, and calibration analysis.

The framework makes four contributions. First, it introduces a three-output architecture (T_{abs} , T_{perc} , I) with a formally established input-space hierarchy. Second, it derives likelihood ratio functions for Bayesian updating from cluster-specific neutral points, reducing free parameters. Third, it identifies and patches a topological vulnerability in naive falsifiability-consistency formulations that would allow perfectly consistent but empirically inert claims to achieve high truth-probability. Fourth, it introduces a relevance-gated situational context mechanism that prevents high-stakes background situations from inflating importance unless the statement itself bears on those stakes.

2 Philosophical Foundation and Core Architecture

2.1 The Three-Output Architecture

The framework produces three scores for any statement s and receiver r :

- $T_{\text{abs}}(s) \in [0,1]$: evidence-weighted truth-probability, receiver-independent. Computed as if evaluated by a highly unbiased and fairly skeptical expert.
- $T_{\text{perc}}(s,r) \in [0,1]$: perceived truth-probability, receiver-dependent.
- $I(s,r) \in [0,1]$: importance score, always receiver-dependent.

The central structural claim is the strict input-space hierarchy. Let I_{abs} , I_{perc} , and I_I denote the input spaces of T_{abs} , T_{perc} , and I respectively:

$$I_{\text{abs}} \subset I_{\text{perc}} \subset I_I$$

Every input that determines T_{abs} also determines T_{perc} and I , and every input that determines T_{perc} also determines I , but not conversely. T_{abs} and T_{perc} are projections of I onto specific axes of the unified scoring space.

2.2 Formal Axiom System

The following fifteen axioms ground the architectural decisions. Note that Axiom 1 has been restated relative to earlier drafts: the framework does not claim to compute a metaphysically objective truth-probability, but rather an evidence-weighted posterior under a specific epistemic stance.

#	Axiom	Statement
1	Evidence-Weighted Truth	For any statement s , there exists an evidence-weighted truth-probability $T_{\text{abs}}(s) \in [0,1]$ representing the degree to which s is supported by available evidence when evaluated by a highly unbiased and fairly skeptical expert. Such an expert defaults to the null hypothesis and requires falsifiable evidence before updating (scientific skepticism), and requires strong cumulative evidence to move substantially from the prior (Bayesian skepticism). These are complementary but distinct epistemic dispositions: the former governs what counts as admissible evidence; the latter governs how much evidence is required.
2	Receiver Dependence of Perception	For any statement s , perceived truth $T_{\text{perc}}(s,r)$ is a function of both s and receiver r . No two receivers are guaranteed to produce identical T_{perc} for the same s .
3	Objective Truth as Perception Limit	As receiver knowledge P approaches 1 and all perceptual distortions vanish, $T_{\text{perc}}(s,r)$ approaches $T_{\text{abs}}(s)$.
4	Importance Undefined Without Receiver	For any statement s , $I(s)$ without a specified receiver r is undefined. Importance is strictly a function $I(s,r)$.
5	Strict Superset Hierarchy	Every input that determines $T_{\text{abs}}(s)$ also determines $T_{\text{perc}}(s,r)$ and $I(s,r)$. Every input that determines $T_{\text{perc}}(s,r)$ also determines $I(s,r)$. The converse does not hold in either case.

6	Evidence Monotonicity	For any statement s , additional independent corroborating evidence never decreases $T_{abs}(s)$. Additional contradicting evidence never increases $T_{abs}(s)$.
7	Bias as Self-Amplifying Corruptor	A source with maximum bias ($B_{eff} \rightarrow 0$) has zero credibility regardless of all other source properties. The credibility damage of bias scales with the degree of bias itself.
8	Falsifiability as Truth Condition	A statement that is in principle unfalsifiable cannot be assigned a non-trivial evidence-weighted truth-probability. Unfalsifiability breaks the Bayesian update engine: if a claim cannot be tested, the likelihood ratio of any evidence observation equals 1, and the posterior never updates from the prior.
9	Self-Report Inferiority	Self-reported signals carry strictly less evidential weight than structurally verifiable signals of the same magnitude.
10	Expertise as Prerequisite	A source with zero domain expertise has zero credibility on statements in that domain regardless of all other source properties.
11	Importance Humanity Principle	Perceptual and receiver factors contribute strictly more to importance than to perceived truth. Evidence-weighted truth contributes strictly less to importance than to perceived truth.
12	Structural Signal Dominance Ordering	Among delivery signals, linguistic register carries strictly more evidential weight than tone, which carries strictly more than expression.
13	Stakes Radius Principle	Among situational signals, the evidential weight of stakes is proportional to the radius of affected parties. Broad societal stakes outweigh speaker stakes, which outweigh background situation.
14	Opposition Superlinearity	The presence of active opposition is a superlinear negative signal on perceived truth. Its evidential weight grows faster than linearly with opposition strength.
15	Statement-Stakes Relevance	Situational stakes may influence perceived truth and importance only to the degree that the statement itself bears informational, causal, operational, evidential, emotional-social, or decision-relevant connection to those stakes. Contextual co-location alone is insufficient for high situational importance.

Table 1: Formal Axiom System

3 Variable Master List

All inputs are normalized to [0,1] unless stated otherwise. Higher values always represent the more credibility-positive or truth-positive direction.

3.1 Cluster 1 — Source Credibility

Symbol	Variable	Description
B = (B1...B4)	Bias vector	Domain-specific bias: B1=political, B2=financial, B3=institutional, B4=motivated reasoning. 0=fully biased, 1=no bias.
B_eff(s)	Effective bias scalar	Statement-weighted collapse: $B_eff(s) = \sum w(s,i) \cdot B_i / \sum w(s,i)$. Uses uniform weights by default (Open Problem 8).
C = (C1...C4)	Expertise vector	C1=formal credentials, C2=peer recognition, C3=years practice, C4=novel contributions track record.
C_eff(s)	Effective expertise scalar	Same weight vector as B_eff: $C_eff(s) = \sum w(s,i) \cdot C_i / \sum w(s,i)$. Note: B_eff and C_eff share the same weight vector w(s,i) — a simplifying assumption acknowledged in Section 9.
D	Rank relative to receiver	Social, institutional, or epistemic standing of source relative to this specific receiver.
E	Stated confidence	Self-reported certainty. Carries less weight than D by Axiom 9.
w(s,i)	Domain relevance weights	Derived from primary and adjacent subjects of statement s. Default: uniform. Formal derivation is Open Problem 8.

Table 2: Cluster 1 — Source Credibility Variables

3.2 Cluster 2 — Statement Properties

Symbol	Variable	Description
F	Novelty relative to field	How surprising this statement is relative to current knowledge in its domain. Used through $F_bal = 4F(1-F)$ in T_abs only.
G	Internal logical consistency	Degree to which the statement contains no internal contradictions. Necessary but insufficient for truth.
H	Falsifiability	Degree to which the statement is in principle testable and disprovable. Tethers the claim to empirical reality.
J	Specificity	Degree to which the statement makes a precise, bounded claim.
k	Falsifiability steepness	$k = \ln(100) = 4.605$, derived so that max of $(1 - e^{-(kH)}) = 0.99$ at $H=1$.

Table 3: Cluster 2 — Statement Properties Variables

3.3 Cluster 3 — Delivery Context

Symbol	Variable	Description
S	Language register	Match between linguistic register and receiver's comprehension and epistemic expectations.

T	Tone	Appropriateness of affective tone to content and context.
U	Posture and expression	Congruence of nonverbal signals with statement. Weakest delivery signal.
V	Time context	Receptivity of temporal setting.
W	Location	Institutional and physical weight of setting.
X	Public vs private setting	Degree to which statement is made in a publicly observable context. X=0 means statement to oneself: Deliv=0 by limit.

Table 4: Cluster 3 — Delivery Context Variables

3.4 Cluster 4 — Situational Context

Symbol	Variable	Description
Y	General situation	Background charge of situational context around the statement's subject at time of delivery.
Z	Speaker stakes	What the speaker risks by making this statement. Double-edged — handled via Z_eff.
Z_eff	Bias-adjusted stakes	$Z_{eff} = Z \times (1 - Z \times (1 - B_{eff}(s)))$. High stakes from a biased source are discounted; high stakes from a clean source remain strong.
AA	Audience composition	Receptivity, expertise, and alignment of the present audience.
AB	Opposing voices	Presence of active counterarguments. 0 = maximum opposition, 1 = no opposition.
AD	Broad stakes	Stakes for all parties beyond the speaker: audience, institutions, society.
Rel(s,c)	Statement-to-stakes relevance	Degree to which statement s itself bears on situational context c. Rel=0 means the statement is merely co-located with the context and does not bear on its stakes. Rel=1 means the statement directly reflects, updates, causes, warns about, requests action about, evidences, or otherwise contributes to the stakes. Intermediate values are scored continuously from the statement and context; no fixed categorical thresholds are assumed.

Table 5: Cluster 4 — Situational Context Variables

3.5 Cluster 5 — Receiver Properties

Novelty N is formalized as a KL divergence between the distribution implied by the statement and the receiver's current belief distribution. Full KL computation requires belief elicitation; the following rubric supports practical scoring:

- $N \approx 0-0.3$: statement is fully consistent with receiver's existing beliefs. No meaningful surprise.
- $N \approx 0.5-0.8$: mild update. Receiver recognizes the claim as somewhat new or reframing.
- $N \approx 1.0-1.5$: meaningful belief update. The claim substantially extends or revises prior beliefs.
- $N \approx 2.0+$: strong divergence. The claim contradicts or substantially expands prior beliefs. Interpolate continuously within and between these anchors.

The sequel study will test whether rater-assigned N values correlate with computed KL on a calibration set of statements.

Symbol	Variable	Description
N	Novelty (KL divergence)	$N = KL(P_s \parallel Q_r)$. P_s = distribution implied by statement; Q_r = receiver's current belief distribution. See Section 5.5 for scoring guidance.
N_perc	Novelty for T_perc	Inverted U: $N_perc = 4.18 \times N_norm^{0.7} \times (1 - N_norm)^{1.5}$, where $N_norm = 1 - e^{-N}$. Peaks near $N_norm \approx 0.32$.
N_imp	Novelty for I(s,r)	Monotonically increasing: $N_imp = 1 - e^{-N(1+N)}$. More novelty always improves importance.
P	Prior knowledge	Receiver's existing domain knowledge. High P pulls T_perc toward T_abs.
Q_rel	Relationship to speaker	Trust, authority, emotional bond, social proximity between receiver and source.
R	Cognitive/emotional state	Degree of emotional engagement. Inverted U — moderate engagement optimal.
AC	Personal consequence	Degree to which the statement's truth would materially change the receiver's world. $AC=0$ collapses Recv to 0.

Table 6: Cluster 5 — Receiver Properties Variables

3.6 Cluster 6 — Corroboration

Symbol	Variable	Description
K1...Kn	Independent sources	n independent corroborating sources. Each contributes one Bayesian update with diminishing likelihood ratios.
L	Established knowledge consistency	Degree to which statement coheres with verified existing knowledge. $x0_L = 0.75$.
M	Physical/empirical evidence	Presence of direct empirical evidence. $x0_M = 0.50$ (floor 0.40 context-dependent).

Table 7: Cluster 6 — Corroboration Variables

4 The Bayesian Update Engine

4.1 Prior — Rolling Average

The prior $P(A)$ for any statement s by a source is the rolling moving average of that source's last $w=10$ evaluated statements:

$$P(A) = (1/w) \times \sum T_abs(s_i)$$

Cold-start anchor: $P(A) = 0.50$ for the first $w-1$ statements, or a domain-specific baseline where available. Past accuracy is encoded as mathematical momentum without the double-counting paradox that would result from including it as both prior and update evidence.

4.2 Likelihood Ratio Function

All cluster scores $x \in [0,1]$ translate to likelihood ratios via:

$$\Lambda_c(x) = \exp((x - x0_c) / (x0_c \times (1 - x0_c)))$$

Where $x0_c$ is the cluster-specific neutral point. This function satisfies: $x = x0_c \rightarrow \Lambda = 1$; $x > x0_c \rightarrow \Lambda > 1$; $x < x0_c \rightarrow \Lambda < 1$; bounded at all $x \in [0,1]$ with no singularities; fully determined by $x0_c$ alone.

Note: this functional form is applied uniformly across all clusters, with only the neutral point varying. Whether cluster-specific steepness parameters would improve predictive validity is an open question (Open Problem 16). The uniform form is a simplifying assumption that reduces free parameters; the sequel study will provide the first empirical test of its adequacy.

4.3 Diminishing Corroboration

To prevent echo chambers and coordinated source inflation, the likelihood ratio of the n -th independent source decays asymptotically:

$$\Lambda_n = 1 + (\Lambda_base - 1) / n^\alpha$$

Where α controls decay speed (Open Problem 5). The 20th corroborating source moves the posterior substantially less than the first.

5 Cluster Scoring Functions

5.1 Cluster 1 — Source Credibility: Cred(s)

$$\text{Cred}(s) = B_{\text{eff}}^{(B_{\text{eff}}^{-0.5})} \times \sqrt{C_{\text{eff}}} \times (4D + E) / 5$$

Structural encoding. $B_{\text{eff}}^{(B_{\text{eff}}^{-0.5})}$: bias self-amplifies its own penalty — a more biased source is punished harder by its own bias. $B_{\text{eff}} \rightarrow 0$ collapses Cred to 0 (Axiom 7). $\sqrt{C_{\text{eff}}}$: expertise enters concavely — the jump from zero to some expertise is large; marginal returns diminish. $C_{\text{eff}} = 0$ collapses Cred to 0 (Axiom 10). $(4D+E)/5$: rank dominates stated confidence 4:1 (Axiom 9). Past accuracy is fully encoded in the rolling prior, eliminating the Bayesian double-counting paradox.

Note on shared weight vector. B_{eff} and C_{eff} use the same domain relevance weights $w(s,i)$. This is a simplifying assumption: in principle, a source could be expert in a subdomain where their bias is also concentrated. Differentiated weight vectors are a possible refinement; the sequel will assess whether this matters empirically.

5.2 Cluster 2 — Statement Properties: Stmt(s)

$$F_{\text{bal}} = 4F(1-F); \text{Stmt}(s) = ((F_{\text{bal}} + J) / 2) \times G \times (1 - e^{(-kH)})$$

Where $k = \ln(100) = 4.605$, derived so the falsifiability term reaches 0.99 at $H=1$.

Structural encoding. $F_{\text{bal}} = 4F(1-F)$: novelty is treated as an inverted-U balance term. It peaks at moderate novelty ($F=0.5$) and penalizes both trivial restatements and extraordinary claims. J enters linearly. G acts as an independent multiplier — internal coherence is necessary but not sufficient. $(1 - e^{(-kH)})$: falsifiability gate. As $H \rightarrow 0$ this term $\rightarrow 0$, collapsing Stmt regardless of G . This closes the vulnerability in the original formulation where $G=1$ and $H=0$ would give $\text{Stmt}=1$, allowing unfalsifiable claims to score at maximum. The separation of G and H reflects that they measure categorically different things: internal coherence versus empirical testability.

5.3 Cluster 3 — Delivery Context: Deliv(s,r)

$$\text{Deliv}(s,r) = W \times (V \times X^{1.1})^{1.1} \times ((3S + 2T + U) / 6)^{0.5}$$

Structural encoding. Location (W) is the base, modulated jointly by time context and public/private setting. $X \rightarrow 0$ collapses Deliv to 0 — a statement made to oneself has no delivery score. Language register dominates tone dominates expression (3:2:1, Axiom 12). The congruence signals enter with square-root concavity.

5.4 Cluster 4 — Situational Context: Sit*(s,r)

$$\text{Rel}(s,c) \in [0,1]$$

$$AD^* = \text{Rel}(s,c) \times AD + (1-\text{Rel}(s,c)) \times x0_AD; \quad Y^* = \text{Rel}(s,c) \times Y + (1-\text{Rel}(s,c)) \times x0_Y; \quad Z_{\text{eff}}^* = \text{Rel}(s,c) \times Z_{\text{eff}} + (1-\text{Rel}(s,c)) \times x0_Z$$

$$\text{Sit}^*(s,r) = AB^{2.1} \times AA^{0.2} \times (2 \times Z_{\text{eff}}^* + 3 \times AD^* + Y^*) / 6$$

Where $Z_{\text{eff}} = Z \times (1 - Z \times (1 - B_{\text{eff}}(s)))$. $\text{Rel}(s,c)$ gates the contextual-stakes variables before they enter the situational-context function.

Structural encoding. $\text{Rel}(s,c)$ is a statement-to-stakes relevance gate. $\text{Rel}=0$ means that the statement is merely co-located with a high-stakes context; AD , Y , and Z_{eff} are pulled back to

their neutral points. $Rel=1$ means that the original situational stakes fully belong to the statement. Intermediate values are not categorical labels. They are continuous judgments based on how strongly the statement itself bears on the surrounding stakes.

Rel may be supported by informational relevance, operational relevance, causal relevance, evidential relevance, harm relevance, need relevance, warning relevance, institutional relevance, or emotional-social relevance. These are scoring cues, not separate variables. A statement may score high Rel through any combination of these cues, and sparse factual reports may receive high Rel when they directly describe damage, need, danger, displacement, infrastructure failure, or decision-relevant change.

Examples. A birthday photograph posted during a hurricane has Rel near 0 with respect to the hurricane context. A general sympathy statement has non-zero emotional-social relevance but lower operational relevance. A report that a road is flooded, a hospital lacks power, or a city is under evacuation has high Rel even if the wording is sparse. The endpoint definitions are fixed; the intermediate scoring remains flexible and rubric-guided.

$AB^{2.1}$ still gives opposing voices a superlinear veto (Axiom 14). $AA^{0.2}$ still makes audience composition weak. Broad stakes still dominate speaker stakes and general situation background when Rel is high (Axiom 13). The new gate prevents ambient background context from being mistaken for statement-specific importance (Axiom 15).

5.5 Cluster 5 — Receiver Properties: $Recv(s,r)$

$$Recv(s,r) = [N_perc^{(1-P)} \times Q_rel \times 4R(1-R)] \times AC^{1.75}$$

Structural encoding. $N_perc^{(1-P)}$: novelty impact decays as prior knowledge grows. Q_rel enters directly. $4R(1-R)$: inverted U for cognitive/emotional state, maximized at $R=0.5$. $AC^{1.75}$: personal consequence superlinear; $AC=0$ collapses $Recv$ to 0.

6 T_{abs}(s) — Evidence-Weighted Truth-Probability

$$T_{abs}(s) = P(A \mid Cred(s), Stmt(s), K1...Kn, L, M)$$

A sequential Bayesian posterior updated over all objective evidence clusters. T_{abs} is receiver-independent in its inputs, but it is not a metaphysically objective truth value — it is the posterior probability that a highly unbiased and fairly skeptical expert would assign given the available evidence. Two distinct sources making identical claims about the same fact will receive different Cred scores and therefore different T_{abs} values. This is a feature, not a flaw: the framework models credibility-adjusted evidence, not raw propositional truth.

6.1 Update Sequence

1. Prior: $P(A)$ = rolling average of source's last $w=10$ evaluated statements. Cold-start: 0.50.
2. Update 1 — Cred(s): $Odds(A|Cred) = \Lambda(Cred) \times Odds(A)$
3. Update 2 — Stmt(s): $Odds = \Lambda(Stmt) \times \text{prior odds}$
4. Updates 3...n+2 — K1...Kn: each independent source applies one update with $\Lambda_n = 1 + (\Lambda_{base}-1)/n^{\alpha}$.
5. Update n+3 — L: consistency with established knowledge. $x0_L = 0.75$.
6. Update n+4 — M: physical evidence. $x0_M = 0.50$ (floor 0.40 context-dependent).

The update sequence is path-independent: order is commutative, only accumulated evidence matters. Conditional independence of cluster evidence given the hypothesis is assumed; relaxing this to model cross-cluster covariance is reserved for future work (Open Problem 3).

7 T_perc(s,r) — Perceived Truth-Probability

$$T_perc(s,r) = P(A \mid T_abs(s), Deliv(s,r), Sit^*(s,r), Recv(s,r))$$

7.1 Update Sequence

7. Prior: $T_abs(s)$ — evidence-weighted posterior as anchor.
8. Update 1 — $Deliv(s,r)$: $Odds = \Lambda(Deliv) \times Odds(T_abs)$
9. Update 2 — $Sit^*(s,r)$: $Odds = \Lambda(Sit^*) \times \text{prior odds}$
10. Update 3 — $Recv(s,r)$: $Odds = \Lambda(Recv) \times \text{prior odds}$

7.2 Key Properties

- T_abs is the anchor — perception deforms it but always starts from it.
- T_perc can exceed or fall below T_abs depending on delivery, relevance-gated situation, and receiver state.
- High P in $Recv(s,r)$ compresses novelty toward zero and pulls T_perc back toward T_abs — knowledgeable receivers converge toward the evidence-weighted estimate (Axiom 3).

8 I(s,r) — Importance Score

$I_context(s,r) = P(\text{importance} \mid \text{Deliv_tilde}, \text{Sit*_tilde}, \text{Recv_tilde})$, prior $P(\text{important}) = 0.50$

$\text{Odds_I} = \text{Odds}(I_context) \times \Lambda_I(T_abs)$, $I(s,r) = \text{Odds_I} / (1 + \text{Odds_I})$

$\Lambda_I = \exp(\text{beta} \times (T_abs^{0.5} - x_0) / (x_0 \times (1 - x_0)))$, $\text{beta} = 0.15$, $x_0 = 0.50$

The prior is flat $P(\text{important}) = 0.50$ — importance does not start from truth, it starts from uncertainty about consequence. T_abs enters as one update among equals, not as the prior anchor. This prevents near-zero T_abs from collapsing importance to zero for false but consequential statements — a critical property for evaluating misinformation. Sit*_tilde enters through the relevance-gated situational context, so ambient background stakes increase importance only when the statement itself bears on those stakes.

8.1 Tilde Transform

$X_tilde_alpha = 1 - e^{-(1 - (1 - X)^{-\alpha})}$. Bounded in $[0, 1]$. $X = 0$ gives $tilde = 0$. $X = 1$ gives $tilde = 1$. Grows fast near high values — small improvements at high delivery/situation/receiver produce large importance jumps.

Term	Alpha	x0_raw	x0_tilde (used in Λ_c)
Deliv_tilde	0.8	0.379	0.372
Sit*_tilde	0.75	0.436	0.415
Recv_tilde	0.7	0.450	0.405

Table 8: Tilde Transform Parameters for Importance Computation

8.2 Key Properties

- Always receiver-dependent — importance without a receiver is undefined (Axiom 4).
- Prior is flat $P = 0.50$. A false statement can be enormously important; truth is one signal among equals, not the anchor.
- T_abs enters via a dampened update ($\text{beta} = 0.15$). Near-zero T_abs gives $\Lambda \approx 0.75$ — a mild negative update rather than a catastrophic penalty. Near-unity T_abs gives $\Lambda \approx 1.35$ — a mild positive update rather than an overwhelming boost. This formally instantiates Axiom 11: T_abs carries strictly less evidential weight on I than on T_perc .
- $N_imp = 1 - e^{-(N(1+N))}$ replaces N_perc in Recv for the I computation. N_imp is monotonically increasing — more novelty always improves importance.

$\text{Rel}(s,c)$ prevents ambient context from dominating importance. High-stakes situations raise I strongly only when the statement bears on those stakes; co-location alone is pulled toward situational neutrality.

9 Cross-Cluster Dependency Map

The framework is an interdependent system, not a linear pipeline. The following dependencies exist between clusters:

- **Cluster 2 → Cluster 1:** $w(s,i)$, the domain relevance weight vector, is derived from the primary and adjacent subjects of statement s and feeds both $B_{\text{eff}}(s)$ and $C_{\text{eff}}(s)$. The same weight vector serves both collapses — a simplifying assumption discussed in Section 5.1.
- **Cluster 1 → Cluster 4:** $B_{\text{eff}}(s)$ feeds $Z_{\text{eff}} = Z \times (1 - Z \times (1 - B_{\text{eff}}))$. **Speaker stakes are discounted by source bias.**

Cluster 4 internal: $\text{Rel}(s,c)$ gates Y , Z_{eff} , and AD by pulling them toward neutral when the statement is co-located with a context but does not bear on that context. Rel is therefore a context-membership correction, not an additional evidence source.

- **Cluster 5 internal:** P modulates the cluster's own output via $N_{\text{perc}}^{(1-P)}$.
- **$T_{\text{abs}} \rightarrow$ Cluster 1 prior:** T_{abs} feeds back into the rolling prior $P(A)$ for future evaluations of the same source.
- **Cluster 5 → T_{perc} and I :** N splits: N_{perc} (inverted U) for T_{perc} ; N_{imp} (monotonic) for I .

10 Neutral Point Table

Every variable has a cluster-specific neutral point $x0_c$ — the score at which the likelihood ratio equals 1 and the prior is unchanged. All values in the table below are principled semantic defaults, not derived values. Their empirical validity — whether they produce rankings that match benchmark targets and human-aligned scoring intuitions — is an open question addressed in the sequel study. The sequel will provide the first external test of whether these defaults hold up across truth, perceived-truth, and importance datasets.

Cluster	Variable	$x0_c$	Semantic Justification
1	B_eff	0.80	Bias at 0.5 is already significantly compromised. High bar for neutrality.
1	C_eff	0.60	Some expertise is the baseline expectation for a source making a claim.
1	D	0.50	Rank is genuinely symmetric — equal rank is truly neutral.
1	E	0.50	Moderate confidence is neutral — neither over nor underconfident.
2	F	0.40	Balanced novelty peaks at moderate surprise; both trivial and extraordinary claims are penalized by $F_bal = 4F(1-F)$.
2	G	0.85	Consistency is a baseline expectation. Half-consistent statements are deeply problematic.
2	H	0.80	High falsifiability is expected. Moderate falsifiability is already epistemically weak.
2	J	0.55	Moderate specificity is acceptable. More symmetric than G or H.
3	S	0.60	Language mismatch is immediately noticeable. Appropriate register is expected.
3	T	0.55	Tone is more forgiving than language. Moderate deviation tolerated.
3	U	0.52	Expression is the weakest signal. Nearly symmetric.
3	V	0.55	Wrong timing is noticeable but not catastrophic.
3	W	0.60	Location carries institutional weight. Appropriate setting is expected.
3	X	0.65	Public setting is expected for credible statements.
4	Y	0.51	General situation is nearly neutral background.
4	Z_eff	0.60	Some stakes expected. High stakes from a clean source is a positive signal.
4	AA	0.60	Receptive audience is expected in normal communication.
4	AB	0.75	Absence of opposition is expected. Even moderate opposition is concerning.
4	AD	0.40	Broad stakes present is a positive signal. Low broad stakes is genuinely neutral.
4	Rel(s,c)	0.50	Moderate statement-to-context relevance is neutral for audit bookkeeping. The endpoint definitions are fixed: Rel=0 means co-location without relevance; Rel=1 means direct relevance to the stakes. Intermediate values remain continuous and rubric-guided, not categorical thresholds.
5	N_perc	0.25	Below the ~32% inverted U peak — novelty below this level is insufficient.
5	P	0.60	Some knowledge expected. Zero knowledge is already problematic.

5	Q_rel	0.50	Stranger to stranger is a genuine baseline.
5	R	0.50	Optimal at 0.5 by design of $4R(1-R)$.
5	AC	0.4	Low consequence is already below neutral — anything touching your world registers.
6	L	0.75	High consistency with established knowledge is strongly expected.
6	M	0.50	Physical evidence is rare. Its absence is neutral; presence is a strong boost.

Table 9: Neutral Point Table. All values are semantic defaults subject to empirical revision in the sequel study.

10.1 Magnitude Normalization

Multiplicative aggregation of variables in $[0,1]$ produces systematic magnitude compression. The standard correction is a monotonic root transformation applied to the composite output before it enters the likelihood ratio function. Root transformations preserve the strict ordering of scores while restoring the composite neutral to its intended range.

Cluster	Formula	Transform	x0 (post-transform)
Cred(s)	As defined	None	0.301
Stmt(s)	As defined	None	0.626
Deliv(s,r)	$\text{sqrt}(\text{Deliv})$	1/2 root	0.379
Sit*(s,r)	$\text{sqrt}(\text{Sit}^*)$	1/2 root	0.436
Recv(s,r)	$\text{cbt}(\text{Recv})$	1/3 root	0.450

Table 10: Normalization Transforms and Post-Transform Neutral Points

11 Limit Definitions and Singularity Handling

Condition	Behavior	Resolution
$B_{\text{eff}} \rightarrow 0$	$\text{Cred} \rightarrow 0$	Limit definition: $\lim(B \rightarrow 0) B^{\wedge} (B^{\wedge} (-0.5)) = 0$ for $B \in [0,1)$
$X \rightarrow 0$	$\text{Deliv} \rightarrow 0$	Limit definition: statement to oneself has no delivery score
$H \rightarrow 0$	$\text{Stmt} \rightarrow 0$	Analytic: $\lim(H \rightarrow 0) (1 - e^{\wedge} (-kH)) = 0$
$R = 0$ or 1	$\text{Recv} \rightarrow 0$	Analytic: $4R(1-R) = 0$ at both extremes
$AC = 0$	$\text{Recv} = 0$	Analytic: $0^{\wedge} 1.75 = 0$
$AB = 0$	$\text{Sit}^* = 0$	Analytic: $0^{\wedge} 2.1 = 0$
$P \rightarrow 0$	Recv volatile	Open Problem 4 — requires careful limit analysis
$\text{Rel}(s,c) = 0$	AD^* , Y^* , and Z_{eff}^* return to neutral	Contextual co-location alone cannot produce high situational importance; the statement must bear on the stakes.

Table 11: Limit Definitions and Singularity Handling

12 Theorems

Theorem 1 — Strict Superset Hierarchy

Statement: $I_{\text{abs}} \subset I_{\text{perc}} \subset I_I$. The scores T_{abs} , T_{perc} , and I are real-valued functions defined on these nested spaces.

Proof sketch: Follows directly from Axiom 5 by construction. I_{abs} contains the input variables of Clusters 1, 2, and 6 only. I_{perc} adds Clusters 3, 4, and 5. I_I adds importance exponents and substitutes N_{imp} for N_{perc} . No input of I_{abs} is absent from I_{perc} or I_I ; no input of I_{perc} is absent from I_I . The converse fails by counterexample — AB is an element of I_{perc} and I_I but not I_{abs} . QED by Axiom 5.

Theorem 2 — Convergence of Perception to Evidence-Weighted Truth

Statement: As $P \rightarrow 1$ and all perceptual inputs approach their neutral points, $T_{\text{perc}}(s,r) \rightarrow T_{\text{abs}}(s)$.

Proof sketch: Under the neutral state N^* (all variables at their $x0_c$ values), every $\Lambda_c = 1$ by definition of $x0_c$. The rolling prior is unchanged. Therefore posterior odds equal prior odds, and $T_{\text{perc}} = T_{\text{abs}}$. The $P \rightarrow 1$ condition is necessary because $N_{\text{perc}}^{(1-P)} \rightarrow 1$ as $P \rightarrow 1$, eliminating novelty's contribution. Formal monotonicity proof deferred — Theorem 6, Open Problem 11.

Theorem 3 — Importance Strict Dominance

Statement: The input space I_I of $I(s,r)$ strictly contains the input space I_{perc} of $T_{\text{perc}}(s,r)$, which strictly contains the input space I_{abs} of $T_{\text{abs}}(s)$.

Proof sketch: The input space of I contains all inputs of T_{perc} , including the relevance-gated situational path, plus the importance exponents and the N_{imp} substitution. Setting those additional elements to recover T_{perc} requires information not present in T_{perc} 's output. Therefore the scoring spaces are strictly nested. Axioms 4, 5, 11, and 15 jointly establish this.

Theorem 4 — Importance Receiver Necessity

Statement: There exists no well-defined $I(s)$ without a specified receiver r .

Proof: $I(s,r)$ depends on $\text{Recv}(s,r)$ which contains AC, Q_{rel} , R, P — all undefined without r . By Axiom 4, importance without a receiver is undefined. QED.

13 Open Problems

#	Problem	Description	Status
1	Variable operationalization	Formal measurement functions for Q_{rel} , AA , $Rel(s,c)$, and remaining intuition-based inputs.	Open
2	Likelihood functions	$P(\text{cluster_score} A)$ and $P(\text{cluster_score} \neg A)$ for each update. Partially resolved via $\Lambda(x)$.	Partially resolved
3	Projection proof	Formally proving T_{abs} and T_{perc} are obtainable as restrictions of I to I_{abs} and I_{perc} respectively.	Open
4	$P \rightarrow 0$ singularity	Behavior of $Recv(s,r)$ as receiver prior knowledge approaches zero.	Open
5	Corroboration decay alpha	Formally defining α in $\Lambda_n = 1+(\Lambda_{base}-1)/n^\alpha$.	Open
6	Steepness constant k	Derived as $k = \ln(100) = 4.605$ from $\max=0.99$ condition.	Resolved
7	Composite neutral points	Resolved via root transformations. See Section 10.	Resolved
8	Weight derivation $w(s,i)$	Formal computation of domain relevance weights. Default: uniform weights assumed for all stress-test cases.	Open — default stated
9	Q_{rel} operationalization	Formal measurement of receiver-speaker relationship strength.	Open
10	Window size w	Empirical justification for default $w=10$ in rolling prior.	Open
11	Knowledge Monotonicity (Theorem 6)	Formally prove T_{perc} monotonically converges toward T_{abs} as P increases.	Open
12	Empirical validation	Testing framework outputs against ground truth datasets of credibility and importance ratings, including independent blind rescoring and inter-rater reliability. Addressed by the empirical stress-test sequence.	Open — empirical sequence
13	N normalization	Validate $N_{norm} = 1-e^{(-N)}$ as appropriate normalization for KL divergence inputs.	Open
14	Temporal dependence of T_{abs}	T_{abs} is formulated as time-independent but evidence quality is time-dependent (e.g. Powell case). Requires temporal indexing $T_{abs}(s,t)$.	Open
15	Population-level perceived truth and importance	Formal definition of $T_{perc,pop}(s)$ and $I_{pop}(s)$ as integrals of receiver-dependent scores over a receiver distribution, including receiver-profile sensitivity analysis.	Open
16	Uniform likelihood ratio form	The Λ_c function uses the same functional form across all clusters, varying only the neutral point. Whether cluster-specific steepness parameters improve predictive validity is untested.	Open — sequel

Table 12: Open Problems

14 Declared Empirical Parameters

The following parameters are acknowledged as empirical rather than axiomatically derived. They represent reasonable defaults pending calibration in the sequel study.

Parameter	Default	Justification and Status
w	10	Rolling prior window size. Empirical default pending study of source track record dynamics (Open Problem 10).
x0_c table	See Section 10	All neutral points are principled semantic defaults, not derived values. Their empirical validity — whether they produce rankings that match benchmark targets under the human-aligned rubric — is an open question addressed in the sequel study.
N_perc exponents	0.7, 1.5	Derived from 30% peak threshold, which is itself an empirical estimate of where novelty begins to hurt perceived truth.
alpha	Undefined	Decay rate of corroboration likelihood ratios (Open Problem 5).
beta	0.15	Dampening factor for T_abs in the I engine. Chosen so near-zero T_abs gives $\Lambda \approx 0.75$ (mild penalty) and near-unity T_abs gives $\Lambda \approx 1.35$ (mild boost). Formally instantiates Axiom 11. Subject to empirical recalibration.
M floor	0.40–0.50	x0_M varies by statement type. Domain-specific calibration required.
w(s,i) default	Uniform	All stress-test cases use uniform domain relevance weights. Formal derivation is Open Problem 8.
Rel(s,c)	Rubric-scored continuous gate	Endpoint definitions are fixed: 0 means co-location without statement-to-stakes relevance; 1 means direct statement-to-stakes relevance. Intermediate values are flexible rubric judgments supported by informational, operational, causal, evidential, harm, need, warning, institutional, or emotional-social relevance cues.

Table 13: Declared Empirical Parameters

15 Stress-Test Cases

Fifteen statements were evaluated using the full framework to verify semantic correctness, correct ranking behaviour, and robustness across four critical edge-case types: (a) false but consequential statements, (b) true but unimportant statements, (c) prestigious sources in the wrong domain, and (d) unfalsifiable claims. The Rel gate is treated as part of the situational context: cases whose stakes are directly carried by the statement preserve their situational force, while cases whose context is merely ambient would be pulled toward situational neutrality.

All cases are documented in full in Appendix A, including variable-by-variable assignments and intermediate composite scores. The summary table below reports final scores only.

#	Statement	T_abs	T_perc	I
1	Powell WMD Address (UN, Feb 2003)	0.617	0.997	0.999
2	Trump inflation claim (2025–26)	0.000126	0.00130	0.938
3	COVID vaccine misinformation tweet (Oct 2022)	0.000700	0.00400	0.798
4	Boring truth: Earth revolves around the Sun	0.99984	0.99959	0.238
5	Climate expert flood warning (72-hour imminent)	0.99998	0.99999	0.994
6	Whistleblower: corporate fraud allegation	0.704	0.925	0.923
7	Prestigious lie: Pauling vitamin C cancer cure	0.335	0.561	0.772
8	5G towers caused COVID-19 (viral tweet, 2020)	0.000250	0.000600	0.524
9	Unfalsifiable platitude: everything happens for a reason	0.00800	0.00900	0.355
10	True rumour: anonymous acquisition tip	0.419	0.642	0.887
11	Retracted scientific paper	0.258	0.397	0.810
12	Eyewitness testimony (single witness, crime scene)	0.536	0.673	0.742
13	Expert consensus: IPCC summary for policymakers	0.99999	0.996	0.965
14	Deathbed confession	0.669	0.781	0.699
15	Medical self-diagnosis: WebMD lupus	0.070	0.078	0.344

Table 14: Stress-Test Summary. Full variable assignments and intermediate scores in Appendix A.

15.1 Semantic Audit

Core ranking triplets. Three ranking triplets hold across the importance scores: (1) Powell (0.999) > Trump (0.938) > COVID tweet (0.798), capturing the gradient from a Secretary of State at the UN to a presidential rally claim to an anonymous social media post. (2) Climate expert (0.994) > Whistleblower (0.923) > Pauling lie (0.772). (3) True Rumour (0.887) > 5G claim (0.524) > Unfalsifiable platitude (0.355).

New cases. Expert consensus (Case 13, $I=0.965$) correctly ranks near the climate expert — high corroboration, high expertise, broad stakes — despite political contestation pulling T_{perc} slightly below T_{abs} . The retracted paper (Case 11, $I=0.810$) ranks between the whistleblower and the vaccine tweet, reflecting that published peer-review provides a substantial credibility floor even after retraction. Eyewitness testimony (Case 12, $I=0.742$) correctly scores moderate-to-high importance — high personal consequence for the receiver but low corroboration keeps T_{abs} at 0.536. The deathbed confession (Case 14, $I=0.699$) rises under the corrected Z_{eff} term because high speaker stakes from a relatively clean source are no longer reversed. The WebMD self-diagnosis (Case 15, $I=0.344$) correctly ranks near the unfalsifiable platitude — very low T_{abs} from near-zero C_{eff} , and low delivery and situational context.

15.2 Critical Edge Cases

(a) False but consequential. Trump ($T_{\text{abs}}=0.000126$, $I=0.938$) and COVID tweet ($T_{\text{abs}}=0.000700$, $I=0.798$) score high importance because the dampened $\Lambda \approx 0.75$ allows massive context to dominate when the statement itself bears on public stakes. Under the relevance-gated architecture, this remains appropriate only when Rel is high: false consequential claims should remain important, but irrelevant statements merely posted during consequential events should not inherit the event's full importance.

(b) True but unimportant. Boring Truth ($T_{\text{abs}}=0.99984$, $I=0.238$) correctly scores low. $T_{\text{abs}}^{0.5}$ near unity gives $\Lambda \approx 1.35$, but zero stakes, zero engagement, and near-zero novelty keep I_{context} low. Truth alone cannot inflate importance — Axiom 11 satisfied.

(c) Prestigious source in wrong domain. Pauling ($T_{\text{abs}}=0.335$, $I=0.772$) correctly scores moderate importance while suppressing truth-probability. $C_{\text{eff}} \approx 0.20$ (wrong domain despite Nobel status) and the inverted-U F_{bal} term penalize the extraordinary medical claim. The desperate receiver's AC remains high, so importance stays substantial.

(d) Unfalsifiable claim. $H=0.02$ collapses Stmt to near zero, cascading to $T_{\text{abs}}=0.008$ and $I=0.355$. Non-zero importance remains because the friend relationship and emotional context give the platitude moderate receiver-relative significance.

15.3 Worked Example: Case 1 (Powell WMD Address)

The Powell case is worked in full below, showing every intermediate computation. This is the most analytically interesting case in the dataset: a credible source making a specific, falsifiable, and eventually false claim in the highest-stakes delivery context available.

Step	Value	Notes
Prior $P(A)$	0.700	Rolling average of Powell's past evaluated statements.
B_{eff}	0.62	State Dept. institutional bias toward war narrative; not fabricating but selectively emphasising.
C_{eff}	0.75	Seasoned diplomat and military professional.

D	0.95	US Secretary of State to UN Security Council.
E	0.95	'I can confirm' — maximum stated confidence.
$\text{Cred}(s) = B_{\text{eff}}^{(B_{\text{eff}}^{(-0.5)})} \times \sqrt{C_{\text{eff}}} \times (4D+E)/5$	0.448	Bias self-amplification: $0.62^{(0.62^{(-0.5)})} = 0.62^{1.27} \approx 0.545$. $\sqrt{0.75} \approx 0.866$. $(4 \times 0.95 + 0.95)/5 = 0.95$. Product ≈ 0.448 .
F	0.55	Mobile biological weapons labs were novel.
G	0.90	Internally coherent narrative with named sites and photos.
H	0.85	UN inspectors could verify on-ground.
J	0.88	Very specific — named sites, vehicle descriptions, satellite imagery.
$F_{\text{bal}} = 4F(1-F)$; $\text{Stmt}(s) = ((F_{\text{bal}}+J)/2) \times G \times (1-e^{(-kH)})$	0.825	$F_{\text{bal}} = 4 \times 0.55 \times 0.45 \approx 0.990$. $(0.990 + 0.88)/2 = 0.935$. $\times 0.90 = 0.842$. $(1-e^{(-4.605 \times 0.85)}) \approx 0.980$. Product ≈ 0.825 .
Cred update → posterior	0.825	$\Lambda(0.448, x_0=0.301) \approx 2.014$. Odds: $(0.700/0.300) \times 2.014 = 4.699$. $P \approx 0.825$ after the credibility update. After the Stmt, L, and M updates, final $T_{\text{abs}} = 0.617$.
K1 (CIA satellite presentation)	neutral	The CIA satellite presentation is treated as part of the same intelligence chain rather than as an independent corroborating source; no additional independent K update is applied.
K2 (NSA intercept presentation)	neutral	The NSA intercept presentation is treated as non-independent with K1; no separate corroboration multiplier is applied.
K3 (UK JIC material)	neutral	The UK JIC material is treated as partially dependent on the same allied intelligence ecosystem; no separate corroboration multiplier is applied in this conservative recomputation.
L (consistency with established knowledge)	0.45	Iraq had historical WMD context, but recent inspection uncertainty and lack of direct confirmation pull established-knowledge consistency below neutral.
M (physical evidence)	0.42	Satellite photos, props, and intercepts are treated as indirect, non-publicly verifiable evidence at the time of evaluation; M remains below the physical-evidence neutral point.
T_{abs} (final)	0.617	Bayesian posterior after Cluster 1, Cluster 2, and conservative Cluster 6 updates under the revised F_{bal} formulation.
$\text{Deliv}_{\text{final}} = \sqrt{\text{Deliv}}$	~ 0.95	$W=0.99$, $X=0.99$, $V=0.95$, $(3 \times 0.97 + 2 \times 0.92 + 0.88)/6 = 0.942$. $\sqrt{}$ transform applied.
$\text{Rel}(s,c)$	0.95	The claim directly bears on the UN war-and-security context; the surrounding stakes are not merely ambient. Rel therefore preserves nearly all of AD, Y, and Z_{eff} in the Sit* computation.

$Sit^*_{final} = \sqrt{Sit^*}$	~0.713	Using corrected Z_{eff} and high Rel because the statement directly concerns the UN security context. Rel preserves the stakes term rather than pulling it toward neutral; $\sqrt{}$ transform gives $Sit^*_{final} \approx 0.713$.
$Recv_{final} = \sqrt[3]{Recv}$	~0.89	N_{perc} ($N=1.5$, $P=0.75 \rightarrow N_{perc}^{(1-0.75)}$ suppressed). $Q_{rel}=0.70$. $4 \times 0.65 \times 0.35 = 0.91$. $AC^{1.75} = 0.99^{1.15} \approx 0.983$. $\sqrt[3]{}$ transform applied.
T_{perc}	0.997	T_{abs} (0.617) as anchor; Deliv, Sit, and Recv all positive updates push perceived truth near ceiling.
$I_{context}$	0.999	Flat prior 0.50; Deliv_tilde, Sit*_tilde, and Recv_tilde remain very high under the revised importance engine.
Λ_T (beta=0.15)	1.187	$\exp(0.15 \times (\sqrt{0.617} - 0.5) / 0.25) \approx 1.187$.
I (final)	0.999	$I_{context}$ odds are multiplied by Λ_T and converted back to probability.

Table 15: Full Worked Example — Case 1, Colin Powell WMD Address

16 Conclusion

This paper has presented a unified mathematical framework for truth-probability and importance-scoring of statements, grounded in fifteen formal axioms, six structured input clusters, a derived likelihood ratio function with cluster-specific neutral points, sequential Bayesian updating throughout, and a relevance-gated situational context mechanism.

The central contributions are: the three-output architecture with formally established input-space hierarchy; the separation of evidence-weighted from perceived truth with T_{abs} as anchor for T_{perc} ; the treatment of importance as the full scoring space of which truth scores are projections; the derivation of the falsifiability steepness constant k from a maximum condition; the identification and patching of a topological vulnerability in naive consistency-falsifiability coupling; the dampened Bayesian update architecture for $I(s,r)$ via $\beta=0.15$; and the $Rel(s,c)$ gate, which prevents ambient contextual stakes from inflating importance when the statement itself does not bear on those stakes.

The framework is both theoretically structured and practically tractable. Its declared empirical parameters — neutral points, weight defaults, β , corroboration decay, and Rel scoring — are stated defaults or rubric-guided judgments, not validated constants. The sequel study, applying the framework to benchmark datasets under explicit scoring protocols, will provide external tests of whether these defaults produce rankings that match human epistemic and importance intuitions, and will identify where the framework needs revision. Sixteen open problems are precisely stated to provide a clear agenda for that work.

Researcher's Note. This research was conducted independently, on free-tier consumer access, without institutional affiliation or funding. Contact: m.amine.belachhab@gmail.com

Appendix A: Stress-Test Case Files

Full variable assignments and intermediate composite scores for all fifteen cases are documented in the accompanying case spreadsheets (case_01 through case_15). This appendix provides case summaries and the key variable choices for Cases 11–15, which are new to this version of the paper.

Case 11 — Retracted Scientific Paper

Statement: *"Hydroxychloroquine is effective as a COVID-19 treatment."* — Published in a peer-reviewed journal, subsequently retracted.

Source: Authors of the Surgisphere/Lancet paper (2020). **Receiver:** General practitioner reading the paper at time of publication.

Key variable choices: C_eff = 0.72 (peer-reviewed publication implies institutional vetting). B_eff = 0.50 (commercial data provider motivation — moderate bias). H = 0.92 (highly falsifiable clinical claim). M = 0.55 (published data — later revealed as fabricated, but not at time of evaluation). L = 0.45 (limited prior evidence, some conflicting signals). Prior P(A) = 0.60 (published peer-reviewed source). AC = 0.90 (treatment decision consequences for patients). AB = 0.65 (some early scepticism in community). T_abs = 0.258 (low corroboration quality, conflicting L, moderate bias, and revised F_bal treatment). I = 0.810 (high receiver consequence, high stakes, published source inflates Deliv and context).

Case 12 — Eyewitness Testimony

Statement: *"I saw the defendant leave the building at 11:47 PM on the night of the incident."*

Source: Single eyewitness, no prior relationship to defendant. **Receiver:** Juror in a criminal trial.

Key variable choices: C_eff = 0.40 (no specialist expertise; human memory reliability is low — Axiom 9 applies: self-report carries less weight). B_eff = 0.70 (no clear motivated bias). D = 0.60 (witness has formal standing in a legal proceeding). H = 0.90 (falsifiable — CCTV, other witnesses could corroborate). G = 0.85 (internally specific and coherent). J = 0.88 (specific time, specific action). L = 0.55 (consistent with possible presence but no confirming evidence). M = 0.30 (single self-report; no physical evidence attached). K = none (single witness). Prior P(A) = 0.50 (cold-start). AC = 0.95 (verdict has major life consequences for juror's decision). T_abs = 0.536 (falsifiable, coherent, but limited by low corroboration and self-report constraints). I = 0.742 (high institutional context of courtroom, high personal consequence for juror).

Case 13 — Expert Consensus: IPCC Summary

Statement: *"Human activities have warmed the atmosphere, ocean, and land. Widespread and rapid changes in the atmosphere, ocean, cryosphere and biosphere have occurred."* — IPCC Sixth Assessment Report Summary for Policymakers.

Source: IPCC (Intergovernmental Panel on Climate Change), representing hundreds of authors.

Receiver: Policy-maker with general scientific literacy.

Key variable choices: C_eff = 0.97 (maximum domain expertise — hundreds of lead authors). B_eff = 0.78 (institutional process designed for balance; some governmental influence on Summary language). D = 0.90 (authoritative international scientific body to policy-maker). H = 0.95 (highly falsifiable empirical claim). G = 0.98 (internally coherent, multi-author review). L = 0.98 (consistent with all major independent observational records). M = 0.95 (satellite, ice core, tidal gauge, atmospheric records — overwhelming physical evidence). K1–K5 (five major

independent assessment bodies). AB = 0.65 (politically contested — meaningful opposition in policy sphere, less so in scientific sphere). Prior P(A) = 0.90. T_abs = 0.99999. I = 0.965 (very high across all dimensions; AB pulls Sit down slightly).

Case 14 — Deathbed Confession

Statement: *"I was the one who set the fire that burned down the warehouse in 1987. I acted alone."*

Source: Person on their deathbed, confessing to a family member. **Receiver:** Family member hearing the confession.

Key variable choices: Z = 0.95 (speaker stakes near-maximum — dying person has nothing to lose and everything to gain from honesty; this is the strongest speaker stakes signal possible). B_eff = 0.85 (no obvious motivated bias — no legal consequences possible). C_eff = 0.50 (first-person claim of personal action; no domain expertise required). D = 0.55 (family member — moderate rank). H = 0.85 (potentially verifiable — arson records, contemporaneous reports exist). G = 0.88 (internally specific). J = 0.80 (named event, named year, named action). L = 0.55 (consistent with known event but no corroborating personal connection prior). M = 0.20 (no physical evidence present in the confession itself). K = none. Prior P(A) = 0.50 (cold-start). X = 0.30 (private conversation). W = 0.30 (private setting). AC = 0.75 (significant emotional and potentially legal consequence for family). T_abs = 0.669. I = 0.699 (high speaker stakes and T_abs are positive but low delivery context and corroboration limit overall importance).

Case 15 — Medical Self-Diagnosis: WebMD

Statement: *"I think I have lupus. I read the symptoms on WebMD and I match all of them."*

Source: Layperson (the receiver themselves) describing their self-diagnosis. **Receiver:** Friend listening to the claim.

Key variable choices: C_eff = 0.05 (near-zero medical expertise — this is the Axiom 10 case most directly). B_eff = 0.55 (health anxiety may bias symptom matching; not malicious). E = 0.65 (moderate stated confidence — 'I think'). D = 0.50 (peer to peer). H = 0.88 (lupus is diagnosable — blood tests, specialist assessment). G = 0.60 (internally coherent as a report of subjective experience, but the inferential step is weak). J = 0.70 (named condition, named symptom matching). F = 0.30 (self-diagnosis via web is common). L = 0.40 (WebMD symptom overlap is notoriously non-specific). M = 0.10 (no medical tests; self-report only — Axiom 9 directly applies). Prior P(A) = 0.50. W = 0.40 (casual setting). X = 0.40 (semi-private). AC = 0.45 (health concern for friend, but they're not the patient). T_abs = 0.070. I = 0.344 (low delivery, low context, near-zero expertise collapses credibility; personal consequence for receiver is moderate at most).

Appendix B: Notation Reference

Symbol	Meaning
T_abs(s)	Evidence-weighted truth-probability of statement s
T_perc(s,r)	Perceived truth-probability of s for receiver r
I(s,r)	Importance score of s for receiver r
P(A)	Prior probability that statement s is true
$\Lambda_c(x)$	Likelihood ratio for cluster c at score x
x0_c	Neutral point of cluster c — score at which $\Lambda = 1$
B_eff(s)	Statement-weighted effective bias scalar
C_eff(s)	Statement-weighted effective expertise scalar
w(s,i)	Domain relevance weight. Default: uniform across all stress-test cases.
Z_eff	Bias-adjusted speaker stakes
N	Raw KL divergence: $KL(P_s \parallel Q_r)$
N_norm	Normalized novelty: $1 - e^{(-N)}$
N_perc	Novelty for T_perc (inverted U): $4.18 \times N_{\text{norm}}^{0.7} \times (1 - N_{\text{norm}})^{1.5}$
N_imp	Novelty for I(s,r) (monotonic): $1 - e^{(-N(1+N))}$
k	Falsifiability steepness constant: $\ln(100) = 4.605$
w	Rolling prior window size: default 10
alpha	Corroboration decay exponent (undefined — Open Problem 5)
beta	Dampening factor for T_abs likelihood ratio in I engine. Default 0.15. Formally instantiates Axiom 11.
Rel(s,c)	Statement-to-stakes relevance gate; degree to which statement s itself bears on situational context c.
AD*	Relevance-adjusted broad stakes: $Rel \times AD + (1 - Rel) \times x0_AD$.
Y*	Relevance-adjusted general situation: $Rel \times Y + (1 - Rel) \times x0_Y$.
Z_eff*	Relevance-adjusted speaker stakes: $Rel \times Z_eff + (1 - Rel) \times x0_Z$.
Sit*(s,r)	Relevance-gated situational context score used in T_perc and I.

Table A1: Notation Reference